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ETB ECU TLC-4 Head Downloading Bench Hardware Specifications

Hardware (Version 2.0)

HMC MM AUTO LTD.

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1. Abbreviations

Abbreviation	Description
ATE	Automatic Test Equipment (Machine)
CAN	Controller Area Network
CCW	Counter Clock wise
CW	Clock wise
DUT	Device Under Test (Production Part)
EOL	End of Line
ETB	Electronic Throttle Body
IAT	Inlet Air Temperature
MAP	Manifold Air Pressure
TBA	Throttle Body Assembly
TBC	To Be Calculated
TBD	To Be Decide
TPS	Throttle Position Sensor
UDS	Unified diagnosis Services
VCC	Voltage at Common Collector(+)
VEE	Voltage at Common Emitter(GND)

2. Scope of the document:

The document describes the hardware requirement for ETB ECU Software Downloading machine (TLC).



ATE

3. Hardware Component

1. Barcode Scanner (Irayple RH3124MG0011E 3000 Series with Cable) -4 Qty
2. PLC (Mitsubishi FX5U-64M) – 1 Qty
3. Vector VN1610 – 4 Qty
4. Advantec PC With Monitor & Keyboard – 1 Set
5. Fixtures & Cables- 4 Qty
6. Power Supply (GWinstek-4323) Quad Output – 1 Qty
7. Advantech PCIE-1812-B – 1 Qty

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3.1 Barcode Scanner

The Barcode Scanner is Irayple RH3124MG0011E 3000 Series with cable, working over Ethernet connection.

Since the Machine has four head 1 scanner is required on each head.



Figure 1

3.2 PLC Mitsubishi FX5U-64 (With 4 Analog input 0-10V DC)

One PLC of Mitsubishi FX5U-64 is required to control the automation part of the downloading bench. Refer the table below for details of each input and output.

The analog inputs will be connected to Vsense1 pin of each DUT to check the power latching operation

Input	Label		Output	Label
X0	PartPresent1		Y0	BATconnect1
X1	PartPresent2		Y1	BATconnect2
X2	PartPresent3		Y2	BATconnect3
X3	PartPresent4		Y3	BATconnect4
X4	Unlock1		Y4	KEY_ON1
X5	Unlock2		Y5	KEY_ON2
X6	Unlock3		Y6	KEY_ON3
X7	Unlock4		Y7	KEY_ON4
X10	Start_1		Y10	CAN_Con1
X11	Start_2		Y11	CAN_Con2
X12	Start_3		Y12	CAN_Con3
X13	Start_4		Y13	CAN_Con4
X14	Pass_1		Y14	Lock1
X15	Pass_2		Y15	Lock2
X16	Pass_3		Y16	Lock3
X17	Pass_4		Y17	Lock4
X20	Fail_1		Y20	PB_Light1
X21	Fail_2		Y21	PB_Light2
X22	Fail_3		Y22	PB_Light3
X23	Fail_4		Y23	PB_Light4
X24			Y24	Pass_LED1
X25			Y25	Pass_LED2
X26			Y26	Pass_LED3



Figure 2

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X27			Y27	Pass_LED4
X30			Y30	Fail_LED1
X31			Y31	Fail_LED2
X32			Y32	Fail_LED3
X33			Y33	Fail_LED4
X34			Y34	
X35			Y35	
X36	Air Pressure OK		Y36	
X37	E_STOP		Y37	

3.3 Vector VN1610

1. CAN Interface card will Connect CANH & CANL line to DUT through Relays operated from Y10 -Y13 of PLC



Figure 3

3.4 Advantech PCIE-1812-B

The Multifunction Card is used to switch signals of DUT using relay matrix. Also will generate PWM signal.



Figure 4

Advantech IPC

IPC-510MB- Advantech 4U 19" Rackmount Industrial System.

400 watt ATX power supply ATX, motherboard- AIMB-705G2 support, LGA1151 7th Gen.

CPU intel i7-6700 with industrial CPU cooler. RAM- 16GB(2*8GB DDR4 U-DIMM)1 TB 2.5" SATA Hard disk.

2 GbE LAN ports. 6 Serial Ports and 6 USB.

Dual Display port DVI/VGA, Including Keyboard Mouse and Display 19 Inch dell/HP display, Windows 10 Pro

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3.6 Fixtures and Cables

Four Fixtures with pneumatic lock, Part Presence probe & Unlock PB). The cable connecting fixture and machine hardware must be shielded and grounded at destination. We recommend to use phoenix cables.

3.7 Programmable Power Supply GWinstek-4323 (Quad channel)

Note: This power supply is used for DUT only; other automation components must be supplied using a separate 24V power supply.

3.8 Machine Structure (Including all electrical components)

The machine structure size must be within 600x 800x520 mm or less if possible. Accommodation for all the hardware components is necessary keeping in mind that adequate access for troubleshooting is available.